

Matrices & Vectors



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and B be the matrix with rows $\langle -1, 4 \rangle$ and $\langle 0, 2 \rangle$. Find $A + B$ and $3A$.
 - 2 Using the same A and B , find $A - 2B$.
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Matrix multiplication

- 3 Using A (rows $\langle 2, 1 \rangle, \langle 3, -1 \rangle$) and B (rows $\langle -1, 4 \rangle, \langle 0, 2 \rangle$) from above, find AB .
 - 4 Let A have rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and let $\mathbf{v} = \langle 2, -3 \rangle$. Find $A\mathbf{v}$.
 - 5 Explain, without computing BA , why AB and BA need not be equal for 2×2 matrices.
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Determinant and inverse

- 6 Find $\det A$ for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and state whether A is invertible.
 - 7 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$.
 - 8 Without computing it, explain why the matrix with rows $\langle 4, 2 \rangle$ and $\langle 6, 3 \rangle$ has no inverse.
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Solving systems with the inverse matrix

- 9 Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.
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Vector magnitude and dot product

- 10 For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{w} = \langle -4, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.
 - 11 What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?
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Angle between vectors

- 12 Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.
- 13 Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.